Project Q&A;: RetailSense AI (Project 4)

This document answers the most frequent questions about the architecture, codebase, and algorithms used in the **RetailSense AI** project.

■ General Platform Questions

Q1: What is the main objective of RetailSense AI?

A: RetailSense AI is an end-to-end retail intelligence platform that ingests raw transaction, pricing, and calendar data (modeled after the M5 Walmart forecasting dataset), processes it through a PySpark Medallion pipeline (Bronze -> Silver -> Gold), trains an XGBoost model to forecast unit sales for the next 30 days, and exposes business analytics via an interactive Streamlit dashboard and a Grok LLM-powered natural language assistant.

Q2: What tech stack is used in the project?

A:

- **Data Processing**: PySpark (Apache Spark 3.5.1)
- **Storage Layer**: Delta Lake (delta-spark 3.1.0)
- **Machine Learning**: XGBoost, Scikit-Learn
- **Backend API**: FastAPI, Uvicorn
- **Frontend Dashboard**: Streamlit, Matplotlib, Seaborn
- **LLM Integration**: Groq/Grok API (via OpenAI SDK wrapper)

■ Data Pipeline Questions

Q3: Why did we unpivot/melt the daily sales data?

A: In the raw M5 dataset, sales are formatted horizontally where each day is a column (e.g. `d_1`, `d_2`, ..., `d_1913`). This wide format is extremely inefficient for SQL filtering, joining, and model training.

By unpivoting (melting) the day columns into a **vertical row format** (with `date` and `sales` columns), we standardize the structure. This enables clean joins with calendar and pricing tables and allows database engine index partitioning on date fields.

Q4: How do the Bronze, Silver, and Gold tables relate?

A:

1. **Bronze**: Stores raw unpivoted DataFrames exactly as ingested.

2. **Silver**: Cleans null values, enforces correct data types (e.g., converting prices to float), and joins sales with prices and calendar attributes.

3. **Gold**: Groups Silver data to build summarized business tables:

- `gold_daily_sales`: Aggregate revenue and unit sales per day.
- `gold_store_performance`: Total revenue, units sold, and average price per store.
- `gold_product_performance`: Total revenue, units sold, and average price per product.
- `gold_state_sales`: Regional sales KPIs.
- `gold_forecasting_input`: Cleaned history containing rolling averages and lags, used to train the model.
- `gold_forecast_results`: The 30-day forecast outputs.

■ Machine Learning Questions

Q5: What features are engineered for the forecasting model?

A:

- **Time-series Lags**: `lag_7` (sales 7 days ago), `lag_14`, `lag_28`.
- **Rolling Statistics**: `rolling_7_avg` and `rolling_30_avg` (smoothed trend).
- **Calendar Features**: `wday` (day of week), `month`, `year`, `is_weekend`, `is_holiday` (snap indicators for states).
- **Pricing**: Current `sell_price` of the item.

Q6: How does the recursive forecasting work?

A:

Because time-series models predict the future based on past lags, we cannot predict $t+2$ directly because the actual sales value for $t+1$ is unknown.

To solve this:

1. The model predicts sales for Day 1 ($t+1$).

2. The predicted value is appended to the historical dataset.

3. The lag features are re-calculated using this prediction.

4. The model predicts Day 2 ($t+2$) using the calculated lags.

5. This loop repeats recursively for 30 steps to construct the full 30-day forecast.

■ API & LLM Questions

Q7: How does the AI Assistant answer user queries?

A:

1. The user asks a question in the chat box (e.g. "Suggest inventory actions for the Foods category").

2. The FastAPI backend endpoint (`/api/chat`) runs SQL queries against the Gold Delta/Parquet tables to calculate current metrics (total sales, top products, worst performing items).

3. The API structures these metrics into a short context prompt.

4. The API queries the Grok/Groq LLM model with the prompt:

```
   You are an expert retail analyst. Here is the current performance summary: [summary].<br/>   User Question: [question]<br/>   Answer the question clearly based on the provided numbers.
```

5. The LLM processes the question using the precise KPIs, returning an accurate, context-aware analysis without hallucinating numbers.